Forecasting the effects of climate change

Ecologists examine changes to animal populations and distribution and apply climate change models to forecast how these will change further in the future, over the course of 5, 10, 50, or more years. In the Arctic, for example, where average temperatures are rising more rapidly than anywhere else, the sea ice is contracting. As a result, polar bears have to travel farther in search of ice where they can catch seals, rest, and mate. The farther they swim, the more energy they burn. As the sea ice declines, the polar bears starve. Scientists monitor their numbers and movements and compare this data with changes in sea ice.

The polar bear plays a vital part in the ecology of the Arctic. As an apex predator and keystone species, it must have access to seals, which are its almost exclusive diet. The number of seals regulates the density of polar bears, while polar bear predation in turn regulates the density and reproductive success of seals.